Cambridge (CIE) IGCSE Maths: Extended

Your notes

Sequences

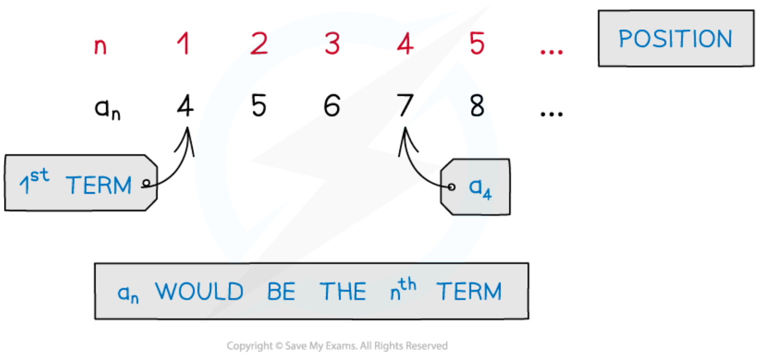
Contents

Your notes

Introduction to Sequences

What are sequences?

- A **sequence** is an ordered set of numbers that follow a **rule**
  - For example 3, 6, 9, 12...
    - The rule is to add 3 each time
- Each number in a sequence is called a **term**
- The **location** of a term within a sequence is called its **position**
  - The letter n is used for position
    - $n = 1$ refers to the **1st term**
    - $n = 2$ refers to the **2nd term**
    - If you do not know its position, you can say the ***n*****th term**
- Another way to show the position of a term is using **subscripts**
  - A general sequence is given by $a_1, a_2, a_3, \ldots$
    - a_1 represents the **1st term**
    - a_2 represents the **2nd term**
    - a_n represents the ***n*****th term**



How do I write out a sequence using a term-to-term rule?

- **Term-to-term rules** tell you how to get the **next term** from the term you are on
  - It is what you do **each time**
  - For example, starting on 4, add 10 each time

- 4, 14, 24, 34, ...

How do I write out a sequence using a position-to-term rule?

- A **position-to-term rule** is an **algebraic expression** in n that lets you find **any term** in the sequence
  - This is also called the ***n* th term formula**
- You need to know what **position** in the sequence you are looking for
  - To get the **1st term**, substitute in $n = 1$
  - To get the **2nd term**, substitute in $n = 2$
- You can jump straight to the 100th term by substituting in $n = 100$
  - You do not need to find all 99 previous terms
- For example, the n th term is $8n + 2$
  - The 1st term is $8 \times 1 + 2 = 10$
  - The 2nd term is $8 \times 2 + 2 = 18$
  - The 100th term is $8 \times 100 + 2 = 802$

How do I know if a value belongs to a sequence?

- If you know the n th term formula, set the value equal to the formula
  - This creates an **equation** to **solve** for n
- For example, a sequence has the n th term formula $8n + 2$
  - Is 98 in the sequence?

$$8n + 2 = 98$$
$$8n = 96$$
$$n = \frac{96}{8}$$
$$n = 12$$

    - It is in the sequence, it is the 12th term
  - Is 124 in the sequence?

$$8n + 2 = 124$$
$$8n = 122$$
$$n = \frac{122}{8}$$
$$n = 15.25$$

    - n is **not** a **whole number**, so it is **not** in the sequence

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Examiner Tips and Tricks

- In the exam, it helps to write the position number (the value of n) above each term in the sequence.

Worked Example

A sequence has the n th term formula $3n + 2$.

(a) Find the first three terms in the sequence.

Substitute $n = 1$, $n = 2$ and $n = 3$ into the formula

$$3 \times 1 + 2 = 5$$

$$3 \times 2 + 2 = 8$$

$$3 \times 3 + 2 = 11$$

5, 8, 11

(b) Find the 80th term.

Substitute $n = 80$ into the formula

$$3 \times 80 + 2$$

The 80th term is 242

(c) Determine whether the number 96 is in the sequence.

Set the formula equal to 96

$$3n + 2 = 96$$

Solve to find n
If n is a whole number, it is a term in the sequence

$$3n = 96 - 2$$
$$3n = 94$$
$$n = \frac{94}{3}$$

94 is not divisible by 3
The nearest possible value is 95 ((95−2) ÷ 3 = 31, the 31st term)

96 is not in the sequence

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Types of Sequences

What are common types of sequences?

- Sequences can follow any rule, but common sequences are
  - **Linear**
    - nth term = $an + b$
  - **Quadratic**
    - nth term = $an^2 + b$
  - **Cubic**
    - nth term = $an^3 + b$
  - **Exponential** (**Geometric**)
    - nth term = $a \times r^{n-1}$
- Sequences may also be formed using common numbers
  - **Prime numbers**
    - 2, 3, 5, 7, 11, ...
  - **Triangular numbers**
    - 1, 3, 6, 10, 15, ...

What is a cubic sequence?

- A **cubic** sequence has an ***n* th term formula** that involves $\boldsymbol{n^3}$
- The **third differences** are **constant** (the same)
  - These are the differences between the second differences
  - For example, 4, 25, 82, 193, 376, 649, ...
    1st Differences: 21, 57, 111, 183, 273, ...

    2nd Differences: 36, 54, 72, 90, ...
    3rd Differences: 18, 18, 18, ...

How do I find the n$^{\text{th}}$ term formula for a cubic sequence?

- The sequence with the n th term formula of $\boldsymbol{n^3}$ is the **cube numbers**
  - 1, 8, 27, 64, 125, ...
    - From $1^3, 2^3, 3^3, 4^3, \ldots$

- Finding the **n th term formula** of other cubic sequences comes from **comparing** them to the cube numbers, n^3
  - 2, 9, 28, 65, 126, ... has the formula $n^3 + 1$
    - Each term is one more than the cube numbers
  - 2, 16, 54, 128, 250, ... has the formula $2n^3$
    - Each term is double a cube number
- You can also use second differences to help find the n th term
  - For the simple **cubic** sequence, n th term = $an^3 + b$
    - where a is $\frac{1}{6}$ of the **third** difference

What is an exponential (geometric) sequence?

- An **exponential (geometric) sequence** is one where you **multiply each term** by the **same number** to get the **next term**
  - E.g. 3, 6, 12, 24, 48, ... is exponential because:
    - terms are **multiplied** by 2 each time
    - 2 is called the **common ratio** (or **constant multiplier**)
    - You can find this by dividing any term by the term immediately before, 6 ÷ 3 or 12 ÷ 6 or 24 ÷ 12 etc
- The **n th term formula** is $\boldsymbol{a \times r^{n-1}}$
  - where a is the first term
  - r is the common ratio
  - n is the position number of the term
    - E.g. 3, 6, 12, 24, 48, ... has $a = 3$ and $r = 2$
    - so the n th term = $3 \times 2^{n-1}$
  - Remember that the power is $n - 1$, not n
- If the common ratio satisfies
  - $r > 1$ then the sequence **increases**
  - $0 < r < 1$ then the sequence **decreases**
    - E.g. $r = \frac{1}{2}$

How can sequences be made harder?

- You may be given a **fraction** with **two different sequences** on the top and bottom

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- E.g. $\frac{3}{1}, \frac{5}{8}, \frac{7}{27}, \frac{9}{64}, \ldots$
  - The **numerators** are the **linear** sequence $2n+1$
  - The **denominators** are the **cube** numbers, n^3
  - So the *n* th term formula is $\frac{2n+1}{n^3}$
- You may be asked to find **combinations** of two different sequences
  - E.g. if sequence *U* is the prime numbers and sequence *V* has the *n* th term formula $4n^2$, find the sequence *U* + *V*
    - *U* = 2, 3, 5, 7, ... and *V* = 4, 16, 36, 64, ...
    - *U* + *V* = (2 + 4), (3 + 16), (5 + 36), (7 + 64), ... = 6, 19, 41, 71, ...
- Other problems involving **setting up and solving equations**
  - This may lead to a pair of **simultaneous equations**

Worked Example

(a) Find the formula for the n^{th} term of the sequence 5, 19, 57, 131, 253, 435, ...

See if the sequence is linear, quadratic or cubic by finding the first, second or third differences

The first differences are

14, 38, 74, 122, 182

These are not constant, so find the second differences

24, 36, 48, 60

These are not constant, so find the third differences

12, 12, 12

These are constant so the sequence is cubic, with n^{th} term formula $an^3 + b$

Method 1

Use the rule that a is $\frac{1}{6}$ of the third difference

$$a = \frac{1}{6} \times 12 = 2$$

This means the n^{th} term formula is $2n^3 + b$
There are different ways to find *b*

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E.g. substitute in $n = 1$ to find the first term from the formula, then make it equal to 5 (the first term in the question)

$$\begin{aligned} 2 \times 1^3 + b &= 5 \\ 2 + b &= 5 \\ b &= 3 \end{aligned}$$

The n^{th} term formula is $2n^3 + 3$

Method 2

Compare 5, 19, 57, 131, 253, 435, ... to the cube numbers 1, 8, 27, 64, 125, ...

Double the cube numbers

2, 16, 54, 128, 250, ...

Add 3

5, 19, 57, 131, 253, ...

The cube numbers (with n^{th} term formula n^3) are doubled then 3 is added

The n^{th} term formula is $2n^3 + 3$

(b) Find the formula for the n^{th} term of the sequence 4, 20, 100, 500, 2500, ...

Seeing if the first, second or third differences are constant does not work

The numbers are increasing very fast, suggesting it could be exponential
Check to see if each term is multiplied by the same number each time

$$4 \times 5 = 20,\ 20 \times 5 = 100,\ 100 \times 5 = 500,\ 500 \times 5 = 2500,\ \ldots$$

Each term is multiplied by 5 (the "common ratio") to get the next, so it is exponential
The n^{th} term formula is $a \times r^{n-1}$ where a is the first term and r is the common ratio

$$a = 4 \text{ and } r = 5$$

The n^{th} term formula is $4 \times 5^{n-1}$

(c) Write down the formula for the n^{th} term of the sequence

$$\frac{5}{4}, \frac{19}{20}, \frac{57}{100}, \frac{131}{500}, \frac{253}{2500}, \ldots$$

This sequence is a fraction formed by dividing the sequence in part (a) by the sequence in part (b)
Divide their n^{th} term formulas

The n^{th} term formula is $\dfrac{2n^3 + 3}{4 \times 5^{n-1}}$

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Worked Example

The first three terms of an exponential sequence are shown below

$$x-1 \qquad 2x \qquad x^2$$

By forming and solving an equation, find the common ratio, r, given that $x \neq 0$.

As this is an exponential sequence, each term is multiplied by the common ratio, r, to get the next term

Consider the first two terms

$$(x-1) \times r = 2x$$
$$r = \frac{2x}{x-1}$$

Considering the next two terms

$$2x \times r = x^2$$
$$r = \frac{x^2}{2x}$$

This is a pair of simultaneous equations
They can be solved by substituting one into the other (replacing r)

$$\frac{2x}{x-1} = \frac{x^2}{2x}$$

Multiply both sides by $2x$

$$\frac{4x^2}{x-1} = x^2$$

Multiply both sides by $x-1$, and expand

$$4x^2 = x^2(x-1)$$
$$4x^2 = x^3 - x^2$$

Subtract $4x^2$ from both sides and factorise

$$0 = x^3 - 5x^2$$
$$0 = x^2(x-5)$$

Solve

$x^2 = 0$ so $x = 0$

or

$$(x-5)=0 \text{ so } x=5$$

You are told $x \neq 0$ so $x=5$

Substitute this into the original sequence, $x-1,\ 2x,\ x^2$

$$(5-1),\ 2(5),\ 5^2$$
$$=4,\ 10,\ 25$$

Find the common ratio r (for example, by dividing a term by its previous term)

$$\frac{10}{4}=2.5$$

$r = 2.5$

Your notes

Linear Sequences

What is a linear sequence?

- A **linear** sequence goes **up** (or **down**) by the **same amount** each time
- This amount is called the **common difference, *d***
  - For example:
    1, 4, 7, 10, 13, ...(adding 3, so $d = 3$)
    15, 10, 5, 0, -5, ... (subtracting 5, so $d = -5$)
- Linear sequences are also called **arithmetic** sequences

How do I find the nth term formula for a linear sequence?

- The formula is **n th term = $dn + b$**
  - ***d*** is the **common difference**
    - The amount it goes up by each time
  - ***b*** is the value **before** the **first** term (sometimes called the **zero** term)
    - Imagine going **backwards**
- For example 5, 7, 9, 11, ...
  - The sequence adds 2 each time
    - $d = 2$
  - Now continue the sequence backwards, from 5, by one term
    - (3), 5, 7, 9, 11, ...
    - $b = 3$
  - So the n th term = $2n + 3$
- For example 15, 10, 5, ...
  - Subtracting 5 each time means $d = -5$
  - Going backwards from 15 gives 15 + 5 = 20
    - (20), 15, 10, 5, ... so $b = 20$
  - The n th term = $-5n + 20$

Worked Example

Find a formula for the n^{th} term of the sequence -7, -3, 1, 5, 9, ...

The n th term is $dn + b$ where d is the common difference and b is the term before the 1st term

The sequence goes up by 4 each time

$$d = 4$$

Continue the sequence backwards by one term (-7−4) to find b

$$(-11), -7, -3, 1, 5, 9, \ldots$$

$$b = -11$$

Substitute $d = 4$ and $b = -11$ into $dn + b$

n^{th} term = $4n - 11$

Your notes

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Quadratic Sequences

What is a quadratic sequence?

- A **quadratic** sequence has an ***n*** **th term formula** that involves $\boldsymbol{n^2}$
- The **second differences** are **constant** (the same)
  - These are the differences between the first differences
  - For example, 3, 9, 19, 33, 51, ...
    1st Differences: 6, 10, 14, 18, ...

    2nd Differences: 4, 4, 4, ...

How do I find the nth term formula for a simple quadratic sequence?

- The sequence with the n th term formula $\boldsymbol{n^2}$ are the **square numbers**
  - 1, 4, 9, 16, 25, 36, 49, ...
    - From $1^2, 2^2, 3^2, 4^2, \ldots$
- Finding the ***n*** **th term formula** comes from **comparing** sequences to the square numbers
  - 2, 5, 10, 17, 26, 37, 50, ... has the formula $n^2 + 1$
    - Each term is 1 more than the square numbers
  - 2, 8, 18, 32, 50, 72, 98, ... has the formula $2n^2$
    - Each term is double a square number
- It may be a simple **combination** of both
  - For example, doubling then adding 1
    - 3, 9, 19, 33, 51, 73, 99, ... has the formula $2n^2 + 1$
- Some sequences are just the square numbers but starting later
  - 16, 25, 36, 49, ... has the formula $(n + 3)^2$
    - Substitute in $n = 1$, $n = 2$, $n = 3$ to see why
- You can also use second differences to help find the n th term
  - For the simple **quadratic** sequence $an^2 + b$
    - a is half of the **second** difference
  - E.g. for the sequence 3, 9, 19, 33, 51, ...

- The second difference is 4, so $a = \frac{1}{2} \times 4$, a = 2
- Compare the sequence $2n^2$, (2, 8, 18, 32, 50, ...) to the original sequence
- The original sequence has the n th term rule $2n^2 + 1$

Examiner Tips and Tricks

- You must learn the square numbers from 1^2 to 15^2

Worked Example

For the sequence 6, 9, 14, 21, 30, ...

(a) Find a formula for the n^{th} term.

Method 1

Find the first and second differences

$$\begin{array}{r} \text{Sequence: } 6,\ 9,\ 14,\ 21,\ 30 \\ \text{First differences: } 3,\ \ 5,\ \ 7,\ \ 9, \dots \\ \text{Second differences: } 2,\ \ 2,\ \ 2, \dots \end{array}$$

The second differences are constant so this must be a quadratic sequence
Compare each term to terms in the sequence n^2 (the square numbers)

$$\begin{array}{r} n^2\text{: } 1,\ 4,\ 9,\ 16,\ 25, \dots \\ \text{Sequence: } 6,\ 9,\ 14,\ 21,\ 30, \dots \end{array}$$

Each term is 5 more than the terms in n^2, so add 5 to n^2

n^{th} term = $n^2 + 5$

Method 2

Find the first and second differences

$$\begin{array}{r} \text{Sequence: } 6,\ 9,\ 14,\ 21,\ 30 \\ \text{First differences: } 3,\ \ 5,\ \ 7,\ \ 9, \dots \\ \text{Second differences: } 2,\ \ 2,\ \ 2, \dots \end{array}$$

Halve the second difference, this is the value of a
Write down the sequence an^2
Compare this to the original sequence

$$\begin{array}{r} 1n^2\text{: } 1,\ 4,\ 9,\ 16,\ 25, \dots \\ \text{Sequence: } 6,\ 9,\ 14,\ 21,\ 30, \dots \end{array}$$

Each term is 5 more than the terms in n^2, so add 5 to n^2

Your notes

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n^{th} **term** $= n^2 + 5$

(b) Hence, find the 20^{th} term of the sequence.

Substitute $n = 20$ into $n^2 + 5$

$$(20)^2 + 5 = 400 + 5$$

The 20^{th} term is 405

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